Siddhant Chauhan

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Summary

B.Tech student in Smart Manufacturing with hands-on experience in CAD design, Finite Element Analysis (FEA), vehicle dynamics, EV powertrain integration, and battery pack design. Experienced in Formula Student race car development, structural validation, thermal analysis, and rapid prototyping through multidisciplinary engineering projects. Seeking core mechanical internships in automotive, electric vehicle, manufacturing, or CAE domains.

Education

PDPM Indian Institute of Information Technology, Design and Manufacturing (IIITDM), Jabalpur **May 2027**

Bachelor of Technology (B.Tech) – Smart Manufacturing

Technical Skills

CAD / CAE	SolidWorks, Fusion 360, ANSYS Workbench
Analysis	Finite Element Analysis (FEA), Stress Analysis, Deformation Analysis, Factor of Safety
Automotive	Vehicle Dynamics, Suspension Design, Steering Geometry
Programming	C++, Python, MATLAB
Embedded & Prototyping	Arduino, Raspberry Pi, Sensor Interfacing
Manufacturing	GD&T, DFM/DFA, Design for Manufacturability

Projects

Formula Student Race Car – Suspension & Structural Design *SolidWorks, ANSYS, Vehicle Dynamics*

- Designed and optimized the suspension system of an FSAE race car to improve handling, stability, and overall vehicle performance.
- Developed lightweight wheel hubs and uprights with a focus on structural integrity and ease of manufacturing.
- Performed load calculations and validated components through finite element analysis by evaluating stress, deformation, and factor of safety.

Retrofitted Electric Bike – Battery Pack Design, Thermal Analysis & Rapid Prototyping *2025*

SolidWorks, ANSYS Workbench, MATLAB

- Converted a KTM Duke chassis into an electric motorcycle by integrating an LFP battery pack, BLDC motor, motor controller, and associated electrical subsystems.
- Engineered the battery pack layout and carried out thermal simulations to ensure safe cell temperatures, improved thermal management, and reliable operation.
- Optimized motor controller parameters through testing and tuning to enhance acceleration, energy efficiency, and overall vehicle performance.

Patent

Automatic Footrest Assembly for Two-Wheelers (Published Patent)

Indian Patent Office

- Designed the complete mechanical architecture of an automated footrest system.
- Developed embedded control logic using Arduino and C++.
- Integrated sensors and actuators with safety-oriented design and validation.

Achievements

- Winner – Smart India Hackathon (SIH) 2025
- 12th Rank – Engineering Design, FSAE Supra 2025

Position of Responsibility

- Vehicle Dynamics Lead – Formula Student (Supra 2026)
- Training and Placement Cell Representative